CHAPTER

2 MRI Reconstruction as an Inverse Problem

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2.1 Inverse problems

An inverse problem arises when we have access to the effects or observed measurements of a certain physical phenomenon and would like to determine its causes. This is the opposite of a forward problem that predicts the effects of certain phenomenon or system given knowledge of their causes. For example, anticipating that you will suffer fever, headache, and fatigue if you have gotten the flu is a forward problem. On the other hand, guessing the specific disease from its symptoms is an inverse problem. From this example, where different diseases may cause the same/similar symptoms, we can notice that inverse problems are more difficult to solve than forward or direct problems.

The difficulty of solving an inverse problem can be characterized by its well-posedness. Specifically, a problem is said to be *well-posed* if it satisfies the three Hadamard conditions: 1) solution existence, 2) solution uniqueness, and 3) solution stability (Fig. 2.1). The first condition is evident: The problem must have a solution. The second condition states that, if a solution exists, it must be unique. The third condition states that the solution should be stable, i.e., small perturbations to the acquired data should produce small perturbations to the solution. If the problem violates one or more of these requirements, it is said to be *ill-posed*. The difficulty is that inverse problems are typically ill-posed. Therefore, a reformulation is needed to find a suitable solution for ill-posed inverse problems. This reformulation often involves including additional requirements or constraints to the solution.[1]

MR reconstruction, and indeed most medical image reconstruction problems, can be treated as an inverse problem since we would like to reconstruct the unknown image from the (noisy) sampled measurements. Specifically, as discussed in Chapter 1, MR reconstruction corresponds to the problem of estimating the transverse macroscopic magnetization (i.e., the image) from the electric potential induced in the receiver coil(s) (i.e., the k-space data). The MR reconstruction problem is typically ill-posed. If the forward model of the reconstruction does not correctly capture the phenomena of the acquisition process, then a solution may not exist (no solution). Since we would (ideally) like to reconstruct the continuous (infinite) distribution of the object's magnetization from a discrete (finite) number of measurements, it is also clear that there are infinite solutions to this problem. For example, the

[1] For more details, the reader is referred to [[22]: "Discrete Inverse Problems: Insight and Algorithms" by Christian Hansen].

Advances in Magnetic Resonance Technology and Applications, Volume 7, ISSN 2666-9099. https://doi.org/10.1016/B978-0-12-822726-8.00011-7

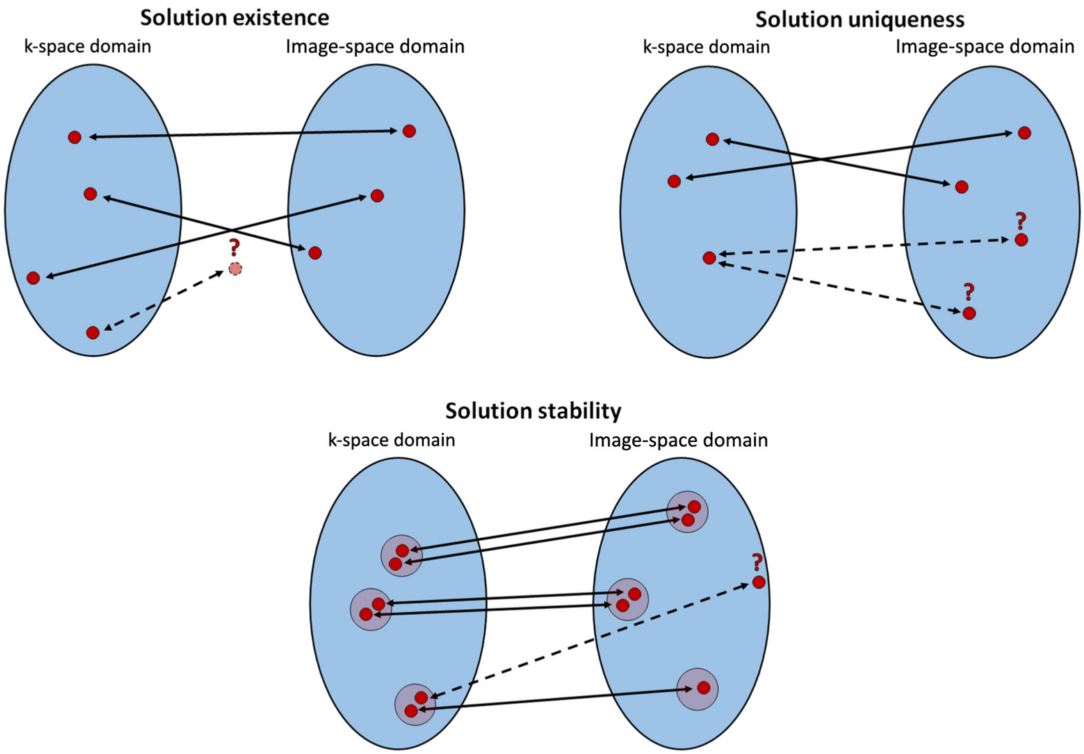

FIGURE 2.1

Representative diagram of the Hadamard conditions that should be satisfied for a well-posed problem. First, a solution must exist according to the underlying mathematical model. Second, the solution must be unique, i.e., a set of k-space measurements should a single solution in the image space domain. Third, the solution must be stable, meaning that small perturbations to the measured data should cause only small perturbations to the image solution.

acquired band-limited signal will impose resolution limitations: there are infinite arrangements of the magnetization at higher resolutions, all of them consistent with the acquired data. If not enough samples are collected (i.e., undersampling), the Fourier transform-based MRI reconstruction problem will be ill-posed and lack a unique solution. This results in a violation of the second Hadamard condition. The third condition, stability, relates to how much perturbations to the acquired data affect the solution. The Fourier transform is in fact sensitive to noise amplification since large perturbations to high-frequency information (e.g., $\sin(kx\pi)$) will amount to small changes to the acquired data $\Delta S(k)$, but can produce arbitrarily large changes in the image $\Delta M(x)$. In other words, $\lim_{k\longrightarrow\infty} \mathcal{F}\{\sin(kx\pi)\} = 0$, which is an instance of the Riemann–Lebesque lemma [1]. This sensitivity to high-frequency perturbations contributes to the instability of the Fourier transform and often manifests itself as noise amplification.

Luckily, the MR reconstruction problem can be reformulated, incorporating additional constraints and regularization terms, to find an appropriate solution to this otherwise ill-posed problem. Hereafter, we will formulate the general MR reconstruction problem, starting with the discretization of the continuous MR signal, and discuss approaches to solve this problem. The formulation and solution to specific MRI applications will be discussed in the remaining chapters of this book.

2.2 Discretization of the MR signal

Chapter 1 introduced the relationship between the measured signal $S(\boldsymbol{k})$ and the transverse magnetization of the underlying object $M_{xy}(\boldsymbol{r})$. If we neglect relaxation effects and receiver coil sensitivity profile, the signal equation is expressed by Principle 6 (Eq. (1.46)):

$$S(\boldsymbol{k}) = \int_V M_{xy}(\boldsymbol{r}) e^{(-2\pi i \boldsymbol{k}\cdot\boldsymbol{r})} d\boldsymbol{r} = \mathfrak{F}\left\{M_{xy}(\boldsymbol{r})\right\} \tag{2.1}$$

where $\mathfrak{F}$ is the Fourier transform and $\boldsymbol{r} = (x, y, z)$ denotes the spatial coordinate; $\boldsymbol{k} = (k_x, k_y, k_z)$ denotes the k-space coordinate and is determined by the magnetic field gradients $\boldsymbol{G}$, $\boldsymbol{k} = \int_0^t \gamma\,(2\pi)^{-1} \times \boldsymbol{G}(t')dt'$. This equation is the forward model that describes the MR acquisition process. As also discussed in Chapter 1, Eq. (2.1) informs that the measured MR signal is given by the Fourier transform of the magnetization; therefore, if $S(\boldsymbol{k})$ is sampled in a continuous (and infinite) fashion, the original magnetization can be obtained via $\mathfrak{F}^{-1}\{S(\boldsymbol{k})\}$. In practice, however, MR acquisition requires the discretization of the acquired (and finite) signal. For example, considerable discretization is required along the phase-encoding direction since data along this dimension is populated along several repetition times (TR), which are limited by scan time. From the full spectrum of $S(\boldsymbol{k})$, only a discrete and band-limited subset of samples $S(\boldsymbol{k_n})$ are acquired in practice.

To investigate the impact of signal discretization on MR reconstruction, we can evaluate $\mathfrak{F}^{-1}\{S(\boldsymbol{k_n})\}$. Starting with the definition of $S(\boldsymbol{k_n})$, the discretized sampling of $S(\boldsymbol{k})$ within a spectral band that is limited by $\boldsymbol{k_{max}}$ (since the signal is finite) is

$$S(\boldsymbol{k_n}) = S(\boldsymbol{k})\, III(\boldsymbol{k}) = S(\boldsymbol{k}) \sum_{-\boldsymbol{k_{max}}}^{+\boldsymbol{k_{max}}} \delta(\boldsymbol{k} - n\boldsymbol{\Delta_k}), \tag{2.2}$$

where III is the comb function, defined as a set of Dirac deltas δ, separated by $\boldsymbol{\Delta_k}$, the space between samples. Rewriting the previous equation as an infinite summation bounded by the *rect* function Π, we arrive at:

$$\mathfrak{F}^{-1}\{S(\boldsymbol{k_n})\} = \mathfrak{F}^{-1}\left\{\Pi(\frac{\boldsymbol{k}}{\boldsymbol{k_{max}}}) S(\boldsymbol{k}) \sum_{-\infty}^{+\infty} \delta(\boldsymbol{k} - n\boldsymbol{\Delta_k})\right\}. \tag{2.3}$$

Following the convolution theorem,

$$\begin{aligned} &\mathfrak{F}^{-1}\left\{\Pi\left(\frac{\boldsymbol{k}}{\boldsymbol{k_{max}}}\right) S(\boldsymbol{k}) \sum_{-\infty}^{+\infty} \delta(\boldsymbol{k} - n\boldsymbol{\Delta_k})\right\} = \\ &= \mathfrak{F}^{-1}\left\{\Pi\left(\frac{\boldsymbol{k}}{2\boldsymbol{k_{max}}}\right)\right\} * \mathfrak{F}^{-1}\{S(\boldsymbol{k})\} * \mathfrak{F}^{-1}\left\{\sum_{-\infty}^{+\infty} \delta(\boldsymbol{k} - n\boldsymbol{\Delta_k})\right\} = \end{aligned} \tag{2.4}$$

$$= 2\boldsymbol{k_{max}} sinc\left(\boldsymbol{r}\boldsymbol{2\pi}\boldsymbol{k_{max}}\right) * M_{xy}\left(\boldsymbol{r}\right) * \left(\boldsymbol{\Delta_k}\right)^{-1} \sum_{-\infty}^{+\infty} \delta\left(\boldsymbol{r} - \frac{n}{\boldsymbol{\Delta_k}}\right).$$

Therefore, the reconstruction from the acquired k-space data can be written as:

$$\mathcal{F}^{-1}\left\{S\left(\boldsymbol{k_n}\right)\right\} = sinc\left(\boldsymbol{r}\boldsymbol{2\pi}\boldsymbol{k_{max}}\right) * \frac{2\boldsymbol{k_{max}}}{\boldsymbol{\Delta_k}} \sum_{-\infty}^{+\infty} M_{xy}\left(\boldsymbol{r} - \frac{n}{\boldsymbol{\Delta_k}}\right). \tag{2.5}$$

The result in Eq. (2.5) is proportional to $\sum_{-\infty}^{+\infty} M_{xy}\left(\boldsymbol{r} - \frac{n}{\boldsymbol{\Delta_k}}\right)$, indicating that a reconstruction from discrete samples will not only reconstruct the original object $M_{xy}(\boldsymbol{r})$, but also infinite replicas $M_{xy}(\boldsymbol{r} - \frac{n}{\boldsymbol{\Delta_k}})$ separated by $\boldsymbol{\Delta_k^{-1}}$ (Fig. 2.2). Therefore, the largest $\boldsymbol{FOV} = \left(\boldsymbol{x_{max}}, \boldsymbol{y_{max}}, \boldsymbol{z_{max}}\right)$ admissible is determined by $\boldsymbol{\Delta_k^{-1}}$ (introduced without derivation in Chapter 1, Eq. (1.52) and Eq. (1.53)). If the object extends beyond this range, then there will be regions where $M_{xy}(\boldsymbol{r})$ and $M_{xy}(\boldsymbol{r} \pm \frac{n}{\boldsymbol{\Delta_k}})$ overlap, creating *aliasing* artifacts. In this case, $\boldsymbol{\Delta_k^{\circ-1}} < \boldsymbol{FOV}$ (where $^{\circ-1}$ denotes the Hadamard inverse), and the reconstruction is characterized as *undersampled* since the sampling of the measurements in k-space does not satisfy the Nyquist-Shannon criteria. If $\boldsymbol{\Delta_k^{\circ-1}} = \boldsymbol{FOV}$, then the reconstruction is characterized as *fully-sampled*, whereas, if $\boldsymbol{\Delta_k^{\circ-1}} > \boldsymbol{FOV}$, then it is referred to as *oversampled*. The acceleration (or undersampling, sometimes also called reduction) factor is commonly used to describe the degree of undersampling: $Acc = N_{acquired}/N_{fully-sampled}$, where $N_{acquired}$ and $N_{fully-sampled}$ are the number of acquired samples (in a given accelerated acquisition) and the number of required samples for a fully-sampled acquisition, respectively. The description so far has focused on a quasiuniform degree of undersampling everywhere in k-space, however some k-space sampling strategies can *oversample* and *undersample* different regions of k-space.

Returning to Eq. (2.5), we note further that the result is convolved with $sinc\left(\boldsymbol{r}\boldsymbol{2\pi}\boldsymbol{k_{max}}\right)$ as a result of having a band-limited signal. This convolution will effectively blur the underlying object M_{xy}, and the resulting spatial resolution $\boldsymbol{\Delta_r}$ can be estimated via

$$\boldsymbol{\Delta_r} = \int_{-\infty}^{+\infty} sinc\left(\boldsymbol{r}\boldsymbol{2\pi}\boldsymbol{k_{max}}\right) d\boldsymbol{r} = \left(2\boldsymbol{k_{max}}\right)^{-1}. \tag{2.6}$$

This relationship was also introduced (without derivation) in Chapter 1, Eqs. (1.54) and (1.55).

Moreover, the convolution of a $sinc$ with any sharp edges present in the image will produce intensity ripples around said edges, known as *Gibbs ringing* (Fig. 2.2). The relationship between k-space sampling and field-of-view ($\boldsymbol{\Delta_k^{-1}} = \boldsymbol{FOV}$), along with the relationship between the resolution and the highest frequency sampled in k-space ($\boldsymbol{\Delta_r} = (2\boldsymbol{k_{max}})^{\circ-1}$), imposes some fundamental trade-offs in MR acquisition. Noting that the total number of acquired samples $\boldsymbol{K_{samples}} = \boldsymbol{2k_{max}} \oslash \boldsymbol{\Delta_k}$ (which is proportional to total scan time) where $\oslash$ denotes Hadamard division, the following relationship can be established:

$$\boldsymbol{K_{samples}} = \boldsymbol{FOV} \oslash \boldsymbol{\Delta_r}. \tag{2.7}$$

Eq. (2.7) notes that there is an inherent trade-off between the field-of-view and resolution. To increase either the FOV or resolution, additional samples must be acquired, extending scan time. The triangular

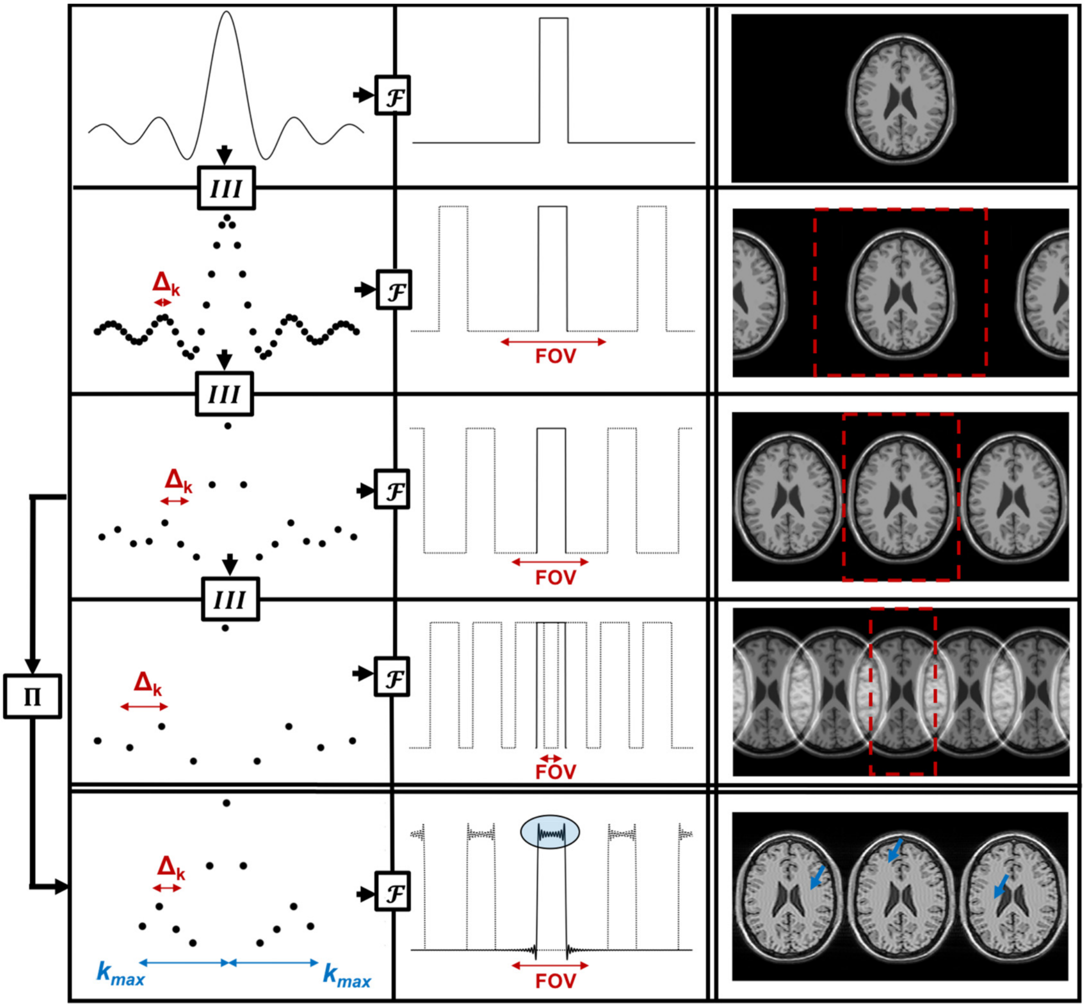

FIGURE 2.2

Practical reconstruction consequences of discrete and band-limited sampling. The first column denotes a set of measurements in k-space domain; the second column contains the corresponding the representation in image-space domain; the third column is an example of MR reconstruction. Discretization of measurements constrains the effective FOV of the reconstruction (red-dashed lines), whereas band-limiting imposes resolution limits and potentially Gibbs ringing (blue arrows).

relationship between resolution, field-of-view, and scan time is a key property of MR and often an important consideration in various MR applications.

Considering that the acquired data is finite, it seems unrealistic that the reconstruction of a continuous (i.e., infinite) object is guaranteed to be correct. As discussed in the previous section, this observation is true, and such a problem is fundamentally ill-posed since we have an infinite number of unknowns to be solved with a finite number of equations. Specifically, there will be an infinite number of solutions to this reconstruction problem, i.e., the solution is not unique. This issue is resolved by assuming that the object $M_{xy}(\boldsymbol{r})$ also lies in a finite-dimensional subspace. Specifically, we will be assuming that the object $M_{xy}(\boldsymbol{r})$ to be reconstructed has finite support (i.e., it is bounded by some $\boldsymbol{FOV}$) and finite resolution (i.e., it is discretized in space, commonly by pixels/voxels). Under these conditions, the following relationship between the continuous ($M_{xy}(\boldsymbol{r})$) and discrete ($M_{xy}(\boldsymbol{r}_n)$) object can be established:

$$M_{xy}(\boldsymbol{r}) = \sum_{n=1}^{N_{pixels}} M_{xy}(\boldsymbol{r}_n)\,\boldsymbol{b}(\boldsymbol{r}-\boldsymbol{r}_n), \tag{2.8}$$

where $\boldsymbol{b}(\boldsymbol{r}-\boldsymbol{r}_n)$ are the chosen basis functions for discretization. The *rect* function is commonly used for this purpose, discretizing the image into pixels, i.e., $\boldsymbol{b}(\boldsymbol{r}) = \Pi\left(\frac{\boldsymbol{r}}{\boldsymbol{\Delta_r}}\right)$.

If the acquisition is *fully-sampled* and the reconstructed image is bounded by $\boldsymbol{FOV}$ (assuming that the object lies on a finite-dimension subspace), then only the original object in Eq. (2.5) remains, i.e., $\sum_{-\infty}^{+\infty} M_{xy}\left(\boldsymbol{r} - \frac{n}{\boldsymbol{\Delta_k}}\right) = M_{xy}(\boldsymbol{r})$. Combining Eq. (2.5) and Eq. (2.8), we obtain

$$\mathcal{F}^{-1}\{S(\boldsymbol{k_n})\} = sinc\left(\frac{\boldsymbol{r}\pi}{\boldsymbol{\Delta_r}}\right) * N_{pixels} \sum_{n=1}^{N_{pixels}} M_{xy}(\boldsymbol{r}_n)\,\Pi\left(\frac{\boldsymbol{r}-\boldsymbol{r}_n}{\boldsymbol{\Delta_r}}\right). \tag{2.9}$$

Eq. (2.9) relates the reconstruction of a discrete, band-limited signal $S(\boldsymbol{k}_n)$ to a discrete image $M_{xy}(\boldsymbol{r}_n)$ blurred by a $sinc$ function if the image is bounded by $\boldsymbol{FOV}$. We have not, up to this point, considered that the acquisition of $S(\boldsymbol{k}_n)$ may be affected by noise, and said noise may affect the reconstructed image.

2.3 MR reconstruction as a linear inverse problem

The discretized (single-channel) signal $S(\boldsymbol{k}_n)$ can be expressed by replacing the continuous object's magnetization ($M_{xy}(\boldsymbol{r})$) in Eq. (2.1) with the relationship derived in Eq. (2.8) and noting the quadrature rule, which results in

$$S(\boldsymbol{k}_n) = \sum_{n=1}^{N_{pixels}} M_{xy}(\boldsymbol{r_n})\,\boldsymbol{b}(\boldsymbol{r}-\boldsymbol{r_n}) e^{(-2\pi i \boldsymbol{k_n}\cdot\boldsymbol{r_n})}\boldsymbol{\Delta_r}. \tag{2.10}$$

This is the discretized forward model of the MR acquisition. Assuming that the pixels are small enough that magnetization changes within a voxel are negligible, a center voxel approximation can be reasonably adopted. Eq. (2.10) can be stated as the linear inverse problem

$$\boldsymbol{s} = \boldsymbol{Ex}, \tag{2.11}$$

where $\boldsymbol{s}$ corresponds to the (vectorized) k-space data, $\boldsymbol{E}$ is the encoding operator that describes the acquisition process, and $\boldsymbol{x}$ is the (vectorized) image. The encoding operator $\boldsymbol{E}$ includes the Fourier transform and other operations that can be included in the forward model, such as the image-space basis $\boldsymbol{b}$, MR physics effects (e.g., relaxation), or the spatial sensitivity profile of the receiving coil array (commonly used for parallel imaging reconstructions, as we will see in Chapter 6). Formulating the MR reconstruction as the linear inverse problem given by Eq. (2.11) is advantageous since we can leverage the many tools of linear algebra to analyze its properties and solve the problem. Further discussion of this topic can be found in the appendix.

As in any measurement system, MRI samples are affected by noise during the measurement process. As discussed in Chapter 1, the noise distribution can (generally) be well-approximated by a simple Gaussian distribution [2]. Therefore, the noise corrupting the MR signal can be modeled by additive white Gaussian noise $\boldsymbol{\varepsilon}$ (following the central limit theorem), and Eq. (2.11) becomes

$$\boldsymbol{s} = \boldsymbol{E}\boldsymbol{x} + \boldsymbol{\varepsilon}. \tag{2.12}$$

Aiming to estimate the image $\boldsymbol{x}$ in the presence of Gaussian noise $\boldsymbol{\varepsilon}$, we turn to Maximum Likelihood Estimation (MLE). Under the MLE framework, we seek to find the best (unbiased) estimate of the image $\hat{\boldsymbol{x}}$ that maximizes the conditional probability that the k-space data $\boldsymbol{s}$ would be measured, in other words,

$$\hat{\boldsymbol{x}} = argmax_{\boldsymbol{x}}\, p\left(\boldsymbol{s}|\boldsymbol{x}\right). \tag{2.13}$$

Assuming that the noise is modeled by an independent, identically distributed (i.i.d.) Gaussian distribution, we can write the noise term as

$$\boldsymbol{\varepsilon} \sim \mathcal{N}\left(z; \mu, \sigma^2\right) = \frac{1}{\sqrt{\pi 2\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}, \tag{2.14}$$

where $\mu = 0$ is the mean and σ is the standard deviation. Since the acquired (single-channel) data $\boldsymbol{s}$ is affected by i.i.d. Gaussian noise (where $\boldsymbol{\varepsilon} = \boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}$), the conditional probability of simultaneously observing all the data points in $\boldsymbol{s}$ is given by

$$p\left(\boldsymbol{s}|\boldsymbol{x}\right) = \prod_{n=1}^{N} \frac{1}{\sqrt{\pi 2\sigma^2}} e^{-\frac{(s_i-(\boldsymbol{E}\boldsymbol{x})_i)^2}{2\sigma^2}}, \tag{2.15}$$

where N is the total number of samples (i.e. $N = K_{samples}$). Noting that log is a monotonically increasing function, the $\hat{\boldsymbol{x}}$ that maximizes $f(x)$ is the same that maximizes $\log f(x)$, i.e.,

$$\hat{\boldsymbol{x}} = argmax_x \log p\left(\boldsymbol{s}|\boldsymbol{x}\right) = N \log \frac{1}{\sqrt{\pi 2\sigma^2}} - \sum_{n=1}^{N} \frac{(s_i - (\boldsymbol{E}\boldsymbol{x})_i)^2}{2\sigma^2}. \tag{2.16}$$

Maximizing a function is equivalent to minimizing the corresponding negative function, i.e., $argmax_x f(x) = argmin_x - f(x)$. Moreover, constants do not affect the computation of the minimum/maximum argument, i.e., $argmin_x f(x) = argmin_x \alpha f(x) + \beta$. Leveraging these two properties, the best image estimate $\hat{\boldsymbol{x}}$ under independent, identically distributed Gaussian noise can be written as:

$$\hat{\boldsymbol{x}} = argmin_x \sum_{n=1}^{N} (s_i - (\boldsymbol{E}\boldsymbol{x})_i)^2. \tag{2.17}$$

In other words, the reconstructed image will be the one that best fits the acquired data under a least squares cost function. This can be written in corresponding vector-matrix notation as

$$\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} \|\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\|_2^2 . \tag{2.18}$$

Note that the least squares formulation results from the noise model, i.e., a different noise model (e.g., Rician, as discussed in Chapter 1) would result in a different formulation. In the following section, we will provide the theoretical background to compute the minimum of this least squares functional to determine the solution of the reconstruction problem, whereas practical algorithms to solve this problem will be discussed in Chapter 3.

2.4 Solution of the MR reconstruction problem

In the previous section, we established that, under i.i.d. Gaussian noise, the linear inverse problem of MR reconstruction amounts to minimizing a least squares cost function:

$$\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} f(\boldsymbol{x}) = argmin_{\boldsymbol{x}} \|\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\|_2^2 . \tag{2.19}$$

The solution image $\hat{\boldsymbol{x}}$ is a minimum argument of $f(\boldsymbol{x})$ if it satisfies the first-order condition

$$\nabla \boldsymbol{f} = \left[\frac{\partial \boldsymbol{f}}{\partial x_1}, \cdots, \frac{\partial \boldsymbol{f}}{\partial x_N}\right]^T = \boldsymbol{0}. \tag{2.20}$$

Let's consider the computation of $\frac{\partial f}{\partial x_n}$, starting by rewriting the cost function

$$f(\boldsymbol{x}) = (\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x})^H (\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}), \tag{2.21}$$

where H is the conjugate transpose (Hermitian transpose). For simplicity, we will use Einstein notation, i.e., the entry in matrix $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ corresponding to the *i-th* row and *k-th* column is written as $C^i_k = A^i_j B^j_k$, where the repeated index implies a summation. In this notation, the least squares cost function can be written as

$$f(\boldsymbol{x}) = \left(s_j - x_i E^i_j\right)\left(s^j - E^j_i x^i\right). \tag{2.22}$$

The partial derivative of the cost function with respect to the *n-th* entry of $\boldsymbol{x}$ is

$$\frac{\partial \boldsymbol{f}}{\partial x_n} = \left(\frac{\partial s_j}{\partial x_n} - \frac{\partial x_i E^i_j}{\partial x_n}\right)\left(s^j - E^j_i x^i\right) + \left(s_j - x_i E^i_j\right)\left(\frac{\partial s^j}{\partial x_n} - \frac{\partial E^j_i x^i}{\partial x_n}\right), \tag{2.23}$$

where the product rule was used. The derivative between two entries of $\boldsymbol{x}$ is zero unless the entry is the same, leading to

$$\begin{aligned}\frac{\partial f}{\partial x_n} &= \left(0-\delta_{in}E^i_j\right)\left(s^j - E^j_i x^i\right) + \left(s_j - x_i E^i_j\right)\left(0 - E^j_i \delta_{in}\right) = \\ &= -E^n_j s^j + E^n_j E^j_i x^i - s_j E^j_n + x_i E^i_j E^j_n = \\ &= -2E^n_j\left(s^j - E^j_i x^i\right),\end{aligned} \tag{2.24}$$

where δ_{in} is the Kronecker delta. Generalizing this last result to every entry in the gradient of the cost function and returning to vector-matrix notation, we obtain:

$$\nabla \boldsymbol{f} = -2\boldsymbol{E}^H\left(\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\right). \tag{2.25}$$

Enforcing the first-order condition,

$$\begin{aligned}\nabla \boldsymbol{f} &= \boldsymbol{0} \\ -2\boldsymbol{E}^H\left(\boldsymbol{s} - \boldsymbol{E}\hat{\boldsymbol{x}}\right) &= \boldsymbol{0} \\ \hat{\boldsymbol{x}} = \left(\boldsymbol{E}^H\boldsymbol{E}\right)^{-1}\boldsymbol{E}^H\boldsymbol{s} &= \boldsymbol{E}^\dagger\boldsymbol{s}.\end{aligned} \tag{2.26}$$

The last line defines the solution of the MR reconstruction, where $\boldsymbol{E}^\dagger$ is known as the Moore–Penrose generalized inverse. If the acquisition is at least *fully-sampled* (meaning $\boldsymbol{\Delta}_{\boldsymbol{k}}^{-1} \geq \boldsymbol{FOV}$, i.e., there are at least as many data points in $\boldsymbol{s}$ as unknowns in $\boldsymbol{x}$) and the noise level is adequate, then we can generally expect a well-posed problem. If a single coil fully sampled Cartesian acquisition is considered, then the forward model simplifies to a Fourier matrix, which has a condition number of 1. In such a case, $\boldsymbol{E} = \boldsymbol{F}$ and a very fast reconstruction is possible since

$$\hat{\boldsymbol{x}} = \left(\boldsymbol{F}^H\boldsymbol{F}\right)^{-1}\boldsymbol{F}^H\boldsymbol{s} = \boldsymbol{F}^{-1}\boldsymbol{s}, \tag{2.27}$$

where $\boldsymbol{F}$ can be efficiently computed using the fast Fourier transform (FFT) [3]. Returning to our sampling considerations, if the data is *under-sampled* ($\boldsymbol{\Delta}_{\boldsymbol{k}}^{-1} < \boldsymbol{FOV}$, there are fewer points in $\boldsymbol{s}$ than unknowns in $\boldsymbol{x}$), then the condition number is $\sim\infty$ and infinitely many solutions exist. Naturally, the second Hadamard condition does not apply in this case. Consequently, Eq. (2.27) is not guaranteed to produce a correct (or even useful) reconstruction.

Since MR data is commonly acquired with an array of coils (multiple channels) with unique spatially varying sensitivities, these can also easily be incorporated into the *fully sampled* reconstruction (improving SNR due to the spatial proximity to the measured object) as $\hat{\boldsymbol{x}} = \left(\boldsymbol{C}^H\boldsymbol{I}\boldsymbol{C}\right)^{-1}\boldsymbol{C}^H\boldsymbol{F}^{-1}\boldsymbol{s}$, where $\boldsymbol{C}$ are the coil sensitivity maps. This is a type of parallel imaging [4–7] reconstruction and is discussed in detail in Chapter 6.

A point of note is that the computation of $\boldsymbol{E}^\dagger$ in the general case requires the inversion of the matrix $\boldsymbol{E}^H\boldsymbol{E}$. In practice, for even small 2D image reconstructions, the size of the matrix $\boldsymbol{E}$ can easily exceed 10,000 × 10,000, meaning that direct matrix inversion is in general not computationally feasible. Since the least squares cost function is convex and differentiable, we can attempt to reach the solution $\hat{\boldsymbol{x}}$ by iteratively searching for a local minimum (which will also be the global minimum). One of the simplest algorithms to find $\hat{\boldsymbol{x}}$ is the gradient descent method. The field of optimization is rich in methods to solve problems of the type $Ax = b$, of which the gradient descent is one the simplest approaches. Another popular (and much more efficient) algorithm to solve this problem is the conjugate gradient. These and other algorithms will be described in detail in Chapter 3.

As discussed earlier, one of the properties that characterize the well-posedness of the MR reconstruction problem is its sensitivity to noise. This can be captured by the *condition number* of the forward model matrix, $\kappa(\boldsymbol{E})$, that provides a theoretical bound for error propagation in $\boldsymbol{s}$ to the solution $\hat{\boldsymbol{x}}$. In other words, the condition number predicts how small perturbations in the measured signal $\boldsymbol{s}$ affect the solution of the inverse problem $\hat{\boldsymbol{x}}$, i.e., a measure of *stability* of the solution. The condition number dictates the following relationship of (relative) error propagation from $\boldsymbol{s}$ to $\boldsymbol{x}$

$$\begin{gathered}\frac{\left\|\boldsymbol{x}-\hat{\boldsymbol{x}}\right\|_2}{\|\boldsymbol{x}\|_2} \leq \kappa\left(\boldsymbol{E}\right)\frac{\|\boldsymbol{\varepsilon}\|_2}{\|\boldsymbol{s}\|_2}\\ \kappa\left(\boldsymbol{E}\right)=\frac{\tau_{max}}{\tau_{min}},\end{gathered} \tag{2.28}$$

where τ_{max} and τ_{min} correspond to the maximum and minimum singular values of $\boldsymbol{E}$. From this definition, it can be seen that $\kappa\left(\boldsymbol{E}\right) \geq 1$. The matrix is considered well-conditioned if the condition number $\kappa\left(\boldsymbol{E}\right) \approx 1$; in this case, the solution is stable to small data perturbations. As the condition number increases, the matrix $\boldsymbol{E}$ becomes increasingly ill-conditioned, meaning that small perturbations of the right-hand side ($\boldsymbol{s}$) can lead to large perturbations of the solution ($\hat{\boldsymbol{x}}$). The condition number κ effectively becomes infinite when $\boldsymbol{E}$ is singular, i.e., the problem is underdetermined. For example, if the acquired data $\boldsymbol{s}$ is *undersampled* (i.e., $\boldsymbol{\Delta}_{\boldsymbol{k}}^{-1} < \boldsymbol{FOV}$), then the reconstruction problem is ill-posed. However, it is still possible to solve ill-posed problems by incorporating additional information in the form of regularization or constraints.

If the reconstruction problem is ill-posed, applying the least squares solution as previously described can produce different artifacts as can be seen in Fig. 2.3. Low levels of noise have a minor effect on the image, however larger noise levels can obscure smaller image structures and lead to an apparent loss of resolution, even though contrast is mostly unaffected. Acquiring only (approximately) half of the k-space, known as the partial Fourier [8,9], can create blurring/smearing artifacts if Hermitian symmetry is not considered. A uniform undersampling pattern (often referred to as Cartesian undersampling) will generate Acc overlapping replicas of the object (often referred to as ghosting), separated by a distance of FOV/Acc. These can be seen in Fig. 2.2 for 1D Cartesian undersampling and in Fig. 2.3 for 2D Cartesian undersampling. Other types of undersampling patterns also generate specific artifacts. Undersampling with a radial trajectory generates streaking artifacts, and undersampling with a spiral trajectory generates spiral-shaped aliasing, as we will see later in Chapter 4. Nonetheless, it is still possible to reconstruct the correct image in all these cases by incorporating additional information into the problem.

2.5 Regularizing the MR reconstruction problem

In the case of an ill-posed MR reconstruction problem (e.g., due to undersampling), the cost function outlined in Eq. (2.18) and its subsequent solution provided in Eq. (2.25) may not produce an adequate

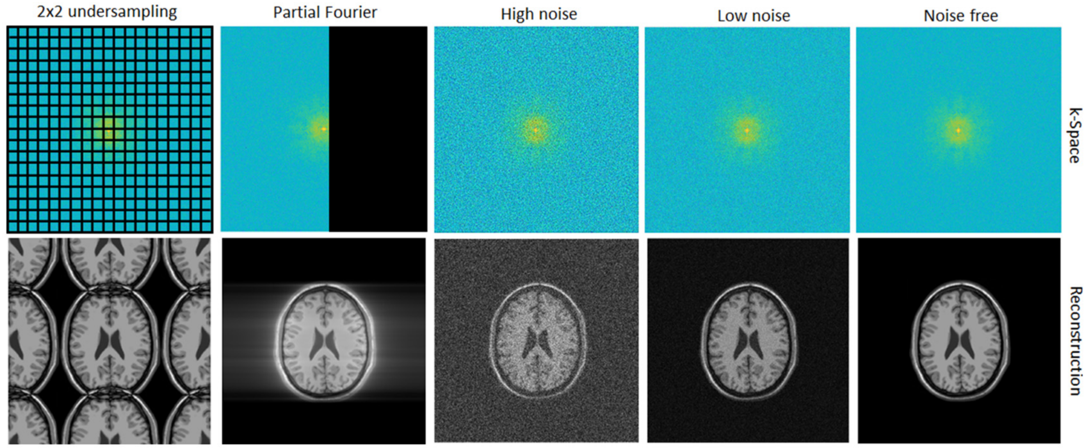

FIGURE 2.3

Reconstructed images from measurements with various sampling properties. Cartesian undersampling leads to aliasing, partial Fourier sampling leads to blurring, and white Gaussian noise perturbations in k-space lead to noise amplification.

image reconstruction of the underlying object. In this case, we can include additional *assumptions* or *prior information* into the cost function to isolate (in the case of multiple solutions) or to stabilize (in the case of unstable solutions) the solution.

These assumptions can be incorporated in the reconstruction via a cost function using a constrained formulation, as shown here:

$$\begin{gathered} \hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} R(\boldsymbol{x}) \\ subject\ to\ \ \|\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\|_2^2 < \varepsilon. \end{gathered} \tag{2.29}$$

This means that, among all feasible solutions, we want to choose the one with the smallest cost function $R(\boldsymbol{x})$ for a given assumption or prior information, while, enforcing consistency between the acquired data and the image we want to reconstruct (i.e., data consistency, $\|\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\|_2^2 < \varepsilon$).

This approach can also be cast as the regularization problem

$$\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} \|\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\|_2^2 + \lambda R(\boldsymbol{x}), \tag{2.30}$$

where the first term enforces consistency between the forward model and the acquired data and the second term $R(\boldsymbol{x})$ adds an assumption or prior information about the image that we want to reconstruct, whereas the regularization parameter λ controls the trade-off between data consistency and faithfulness to the optimization criterion.

For example, say we have an estimate of the image $\boldsymbol{x}_0$ (e.g., obtained from a previous examination of the same patient). We could leverage this prior information by stating that the reconstructed image should be similar (in the least squares sense) to $\boldsymbol{x}_0$, corresponding to the following constrained problem:

$$\begin{gathered}\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} \left\|\boldsymbol{x}-\boldsymbol{x}_0\right\|_2^2 \\ subject\ to\ \left\|\boldsymbol{s}-\boldsymbol{E}\boldsymbol{x}\right\|_2^2 < \varepsilon.\end{gathered} \tag{2.31}$$

For purposes of optimization, it can be more practical to formulate Eq. (2.31) as an unconstrained regularized problem, i.e.,

$$\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} f\left(\boldsymbol{x}\right) = argmin_{\boldsymbol{x}} \left\|\boldsymbol{s}-\boldsymbol{E}\boldsymbol{x}\right\|_2^2 + \lambda\left\|\boldsymbol{x}-\boldsymbol{x}_0\right\|_2^2, \tag{2.32}$$

where the regularization term is $\left\|\boldsymbol{x}-\boldsymbol{x}_0\right\|_2^2$ in this case. For an appropriate choice of λ, Eqs. (2.31) and (2.32) have the same solution. This specific type of regularization is known as Tikhonov regularization [10]. The choice of an L_2-norm ($\|.\|_2$) for the regularizer is practical because it will still lead to a linear problem in $\boldsymbol{x}$ with a closed-form solution. Following the same analysis as outlined in Eqs. (2.22)–(2.26), it can be shown that:

$$\begin{gathered}\frac{\partial f}{\partial x_n} = -2E_j^n\left(s^j - E_i^j x^i\right) + 2\lambda I_j^n I_i^j\left(x^i - x_0^i\right) \\ \nabla f = -2\boldsymbol{E}^H\left(\boldsymbol{s}-\boldsymbol{E}\boldsymbol{x}\right) + 2\lambda\boldsymbol{I}^H\boldsymbol{I}\left(\boldsymbol{x}-\boldsymbol{x}_0\right),\end{gathered} \tag{2.33}$$

where $\boldsymbol{I}$ corresponds to the identity matrix. Setting $\nabla f = \boldsymbol{0}$ leads to the following solution of the Tikhonov regularized problem

$$\hat{\boldsymbol{x}} = \left(\boldsymbol{E}^H\boldsymbol{E} + \lambda\boldsymbol{I}\right)^{-1}\left(\boldsymbol{E}^H\boldsymbol{s} + \lambda\boldsymbol{x}_0\right). \tag{2.34}$$

Increasing λ will make the solution further insensitive to the noise affecting the measurement; however, increasing λ will also bias the solution towards $\boldsymbol{x}_0$, potentially producing a solution $\hat{\boldsymbol{x}}$ that is not consistent with the acquired data. The latter scenario is important to note since regularization can potentially suppress noise and image features within the noise level alike. Let's consider an example where we have a low resolution prior $\boldsymbol{x}_0$ of the image and we wish to incorporate it as regularization in a noisy, undersampled reconstruction (Fig. 2.4). The reconstruction will be under-regularized if λ is too low, leading to similar quality as in the absence of regularization, presenting residual aliasing and noise amplification. In this case, the term $\left\|\boldsymbol{s}-\boldsymbol{E}\boldsymbol{x}\right\|_2^2$ in Eq. (2.32) dominates the cost function. On the other extreme, we have over-regularization (λ is too high); some aliasing and noise are suppressed, however resolution loss is instead observed as the reconstruction converges to the prior. In this case, the term $\lambda\left\|\boldsymbol{x}-\boldsymbol{x}_0\right\|_2^2$ dominates the cost function in Eq. (2.32). The ideal choice of λ achieves a good balance between both terms, allowing aliasing/noise reduction without compromising fidelity to the ground-truth image (Fig. 2.4).

The choice of the regularizer when some prior estimate $\boldsymbol{x}_0$ of the image is known is not limited to this example; any useful linear operator $\boldsymbol{L}$ can also be included in the Tikhonov regularization term. In that case its corresponding solution is given by

$$\begin{gathered}\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} \left\|\boldsymbol{s}-\boldsymbol{E}\boldsymbol{x}\right\|_2^2 + \lambda\left\|\boldsymbol{L}\boldsymbol{x}-\boldsymbol{x}_0\right\|_2^2 \\ \hat{\boldsymbol{x}} = \left(\boldsymbol{E}^H\boldsymbol{E} + \lambda\boldsymbol{L}^H\boldsymbol{L}\right)^{-1}\left(\boldsymbol{E}^H\boldsymbol{s} + \lambda\boldsymbol{L}^H\boldsymbol{x}_0\right).\end{gathered} \tag{2.35}$$

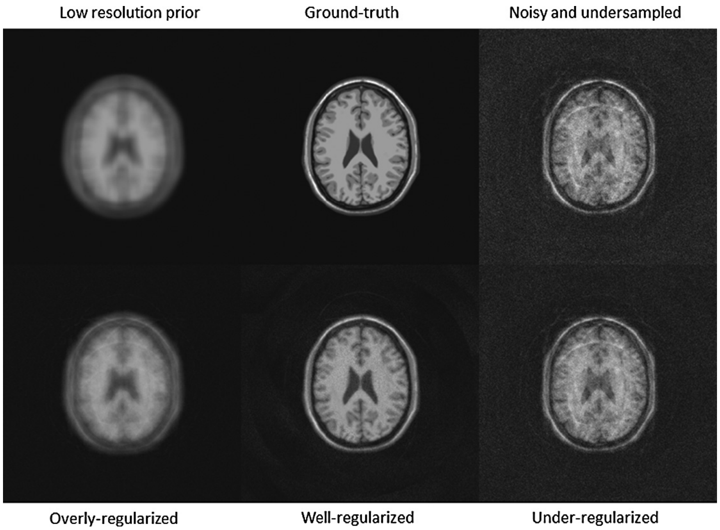

FIGURE 2.4

Example of a regularized reconstruction using a low resolution image as prior information. Image artifacts arise due to undersampling and noisy measurements; however, these can be adequately suppressed, granted the regularization strength is correctly set.

Furthermore, the regularization term does not have to necessarily feature a L_2-norm, for example, compressed sensing [11–14] employs an L_1-norm (as discussed in Chapter 8 in greater detail). Minimizing the L_1-norm of some vector $\boldsymbol{x}$ corresponds to enforcing the prior information that $\boldsymbol{x}$ should be sparse, i.e., have only a few nonzero elements. While there are only a few MR applications where the image itself is sparse, the sparsity constraint can be enforced in some domain (other than the image space) where the transformed $\boldsymbol{x}$ becomes sparse. The general formulation for L_1 regularization is written as

$$\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} f(\boldsymbol{x}) = argmin_{\boldsymbol{x}} \|\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\|_2^2 + \lambda \|\boldsymbol{\Phi}\boldsymbol{x}\|_1, \tag{2.36}$$

where $\boldsymbol{\Phi}$ is referred to as a sparsifying transform (e.g., finite differences or a wavelet transform), i.e., some linear operator that projects the nonsparse image $\boldsymbol{x}$ into another domain where $\boldsymbol{\Phi}\boldsymbol{x}$ is sparse. Computing the gradient of this cost function requires evaluating $\nabla \|\boldsymbol{\Phi}\boldsymbol{x}\|_1$ and consequently $\frac{\partial |x_i|}{\partial x_n}$, which is not well defined. Specifically, the derivative of $|x|$ approaching from 0^- is different if approaching from 0^+ (-1 and $+1$, respectively). One approach is to "round out" the "kink" at $x = 0$ by redefining $|x| \approx \sqrt{x^2 + \epsilon}$. Noting that the partial derivative $\frac{\partial |x_i|}{\partial x_n} = \delta_{in} \frac{x_i}{|x_n|}$ and also that L_1-norm $\|\boldsymbol{\Phi}\boldsymbol{x}\|_1 = \left|\Phi_i^j x^i\right|$, we can compute the partial derivative of the L_1-norm

$$\frac{\partial \left|\Phi_i^j x^i\right|}{\partial x_n} = \Phi^i_{\ j} \delta_{in} \frac{\Phi_i^j x^i}{\left|\Phi_n^j x^n\right|} = \frac{\Phi_j^n \Phi_n^j x^n}{\sqrt{x_n \Phi_j^n \Phi_n^j x^n + \epsilon}}. \tag{2.37}$$

We can then evaluate the gradient of the sparsity-regularized reconstruction

$$\frac{\partial \boldsymbol{f}}{\partial x_n} = -2E_j^n \left(s^j - E_i^j x^i\right) + 2\lambda \frac{\Phi_j^n \Phi_n^j x^n}{\sqrt{x_n \Phi_j^n \Phi_n^j x^n + \epsilon}} \qquad (2.38)$$
$$\nabla \boldsymbol{f} = -2\boldsymbol{E}^H \left(\boldsymbol{s} - \boldsymbol{E}\boldsymbol{x}\right) + \boldsymbol{2}\lambda \boldsymbol{\Phi}^H \boldsymbol{W} \boldsymbol{\Phi} \boldsymbol{x},$$

where $W_n^n = \left(x_n \Phi_j^n \Phi_n^j x^n + \epsilon\right)^{-1/2}$ is a diagonal matrix that includes a term with $x^{-1/2}$. As we can see, the L_1-norm regularization creates a problem that is no longer linear in $\boldsymbol{x}$. Consequently, algorithms designed to solve linear problems will not be suitable for this optimization process and more advanced nonlinear methods must be employed. Suitable algorithms to solve this problem will be discussed in Chapter 3.

Finally, we note that many other forms of regularization are also possible, and a plethora of MR reconstruction methods exist that incorporate additional information into the process to improve performance. The main strategies are explored in Chapters 6–11.

2.6 Nonlinear inverse problems in MR

All reconstruction problems considered up to this point assume a known linear forward model, resulting in a linear inverse problem. However, there are applications in MR where the forward model $\boldsymbol{E}$ depends nonlinearly on parameters $\boldsymbol{\theta}$ that need to be estimated, in addition to the image pixels in $\boldsymbol{x}$. These cases lead to nonlinear inverse problems that can be severely underdetermined due to the increased number of unknowns. The general problem may be stated as

$$\left\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{\theta}}\right\} = argmin_{\boldsymbol{x},\boldsymbol{\theta}}\, g\left(\boldsymbol{x}, \boldsymbol{\theta}\right) = \qquad (2.39)$$
$$= argmin_{\boldsymbol{x},\boldsymbol{\theta}} \left\|\boldsymbol{s} - \boldsymbol{E}(\boldsymbol{\theta})\boldsymbol{x}\right\|_2^2 + \lambda_x \boldsymbol{R}_{\boldsymbol{x}}\left(\boldsymbol{x}\right) + \lambda_{\boldsymbol{\theta}} \boldsymbol{R}_{\boldsymbol{\theta}}\left(\boldsymbol{\theta}\right).$$

Simultaneously, while estimating the image and the forward model can lead to improved results, it results in a more challenging problem than previously discussed in Section 2.3. Evaluating the partial derivatives $\partial g/\partial \boldsymbol{x}$ and $\partial g/\partial \boldsymbol{\theta}$ to identify potential minima reveals a nonlinear solution on $\left\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{\theta}}\right\}$ that is not immediately easy to compute, nor is it necessarily unique. This function may admit multiple (possibly local) minima, and consequently the converged solution is dependent on the initial conditions of the chosen algorithm. Furthermore, even when the data acquisition is fully-sampled, the corresponding nonlinear problem can be underdetermined, as $K_{samples} < N_{pixels} + \theta_{parameters}$. Despite all these challenges, several nonlinear MR forward models have been developed in recent years to avoid assumptions in the forward model, increase robustness to errors in the model, or directly estimate parameters of interest that are embedded in the model. General strategies to overcome these challenges include parametrization to reduce the number of unknowns, splitting the reconstruction into multiple subproblems and linearizing the model around a neighborhood of existing solutions. We will briefly discuss three examples of nonlinear problems in MR: parallel imaging, motion estimation/correction, and MR parameter reconstruction. Detailed analysis of various nonlinear models used in MR can be found in Chapters 12–16.

2.6.1 Nonlinear parallel imaging

A common *image-based* formulation of parallel imaging includes the coil sensitivities $\boldsymbol{C}$ (complex, block diagonal matrix) into the forward model as $\boldsymbol{E} = \boldsymbol{BFC}$, where $\boldsymbol{B}$ is a k-space sampling operator (logical, diagonal sparse matrix). The coil sensitivities can be estimated a priori from calibration data; in that case, the following parallel imaging reconstruction becomes a simple linear inverse problem. Errors in the initial estimation of $\boldsymbol{C}$ can propagate into reconstruction errors in $\boldsymbol{x}$. Recent methods [15,16] have been proposed to simultaneously solve for the image $\boldsymbol{x}$ and the coil maps $\boldsymbol{C}$:

$$\left\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{C}}\right\} = argmin_{\boldsymbol{x},\boldsymbol{C}} \|\boldsymbol{s} - \boldsymbol{E}(\boldsymbol{C})\boldsymbol{x}\|_2^2 + \sum_i \lambda_i \boldsymbol{R_i}(.). \tag{2.40}$$

The operator $\boldsymbol{C} = [\boldsymbol{C_1}, \ldots, \boldsymbol{C_s}]$ is a block-diagonal matrix, where the nonzero entries to be estimated are $c_{r,s}$: one parameter per image location r and per coil channel s. Consequently, even for *fully sampled* data, this problem will be underdetermined since $K_{samples} < N_{pixels} + N_{coils}N_{pixels}$ ($K_{samples} = N_{coils}N_{pixels}$ for *fully sampled* data). Furthermore, the solution is nonlinear for the parameters of interest, requiring more complex optimization algorithms. One approach [15] to this problem is to employ Gauss–Newton algorithms. The strategy here to linearize the problem around the current solution at the *i-th* iteration $\boldsymbol{y}^{[i]}$ and to solve for the small increment to the solution $d\boldsymbol{y}$. Reformulating the problem as $\boldsymbol{E}(\boldsymbol{y}) = \boldsymbol{s}$, where $\boldsymbol{y} = \left[x_1, \ldots, x_r, c_{1,1}, \ldots, c_{r,s}\right] = [\boldsymbol{x}, \boldsymbol{C_1}, \ldots, \boldsymbol{C_s}]$, the forward model can be approximated by the first-order Taylor expansion

$$\boldsymbol{E}\left(\boldsymbol{y}^{[i]} + d\boldsymbol{y}\right) \approx \boldsymbol{E}\left(\boldsymbol{y}^{[i]}\right) + \mathcal{D}\boldsymbol{E}\left(\boldsymbol{y}^{[i]}\right) d\boldsymbol{y} = \boldsymbol{s}, \tag{2.41}$$

where $\mathcal{D}$ is the Jacobian. We can then formulate an approximate linearized inverse problem on $d\boldsymbol{y}$ as

$$d\hat{\boldsymbol{y}}^{[i+1]} = argmin_{d\boldsymbol{y}} \left\|\boldsymbol{s} - \boldsymbol{E}\left(\boldsymbol{y}^{[i]}\right) - \mathcal{D}\boldsymbol{E}\left(\boldsymbol{y}^{[i]}\right) d\boldsymbol{y}\right\|_2^2 + \lambda \boldsymbol{R}(d\boldsymbol{y}), \tag{2.42}$$

for which there are efficient solvers like the conjugate gradient. Iteratively solving for $d\boldsymbol{y}$ and updating $\boldsymbol{y}^{[i+1]} = \boldsymbol{y}^{[i]} + d\boldsymbol{y}^{[i]}$ will generally converge to a solution, however, the final solution of $\hat{\boldsymbol{y}}$ can vary depending on the choice of the initial guess $\boldsymbol{y}^{[0]}$. Note that, before solving the linear inverse problem in Eq. (2.42), the partial derivatives of the encoding operator must be computed, requiring additional computations. Considering $\boldsymbol{E}(\boldsymbol{y}) = [\boldsymbol{E}_1(\boldsymbol{y}), \ldots, \boldsymbol{E}_s(\boldsymbol{y})]^T$, $\partial \boldsymbol{E}_s\left(\boldsymbol{y}^{[i]}\right)/\partial \boldsymbol{x} = \boldsymbol{BFC_s}$ and $\partial \boldsymbol{E}_s\left(\boldsymbol{y}^{[i]}\right)/\partial \boldsymbol{C_s} = \boldsymbol{BF}\boldsymbol{y}^{[i]}$ must be evaluated every time $\boldsymbol{y}$ is updated.

Another approach to tackle the problem of nonlinear parallel imaging is to parametrize the coil sensitivities in order to reduce the number of unknowns, as proposed in [16]. Coil sensitivities are known to be smooth, and therefore can be parametrized by low-order polynomials such as $c_{r,s} = \sum_k^P \alpha_{k,s} r^{\langle k \rangle}$, where P is the order of the polynomial (and $.^{\langle k \rangle}$ denotes the exponential). Instead of solving for $c_{r,s}$ we can solve for $\alpha_{i,s}$, which significantly reduces the number of unknowns since $P \ll N_{coils}N_{pixels}$. A further property of this formulation is that appropriate selection of P will inherently enforce a smooth solution. The corresponding joint minimization problem then becomes

$$\left\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{\theta}}\right\} = argmin_{\boldsymbol{x},\boldsymbol{a}} \|\boldsymbol{s} - \boldsymbol{BFC}(\boldsymbol{\theta})\boldsymbol{x}\|_2^2, \tag{2.43}$$

where $\boldsymbol{\theta}$ contains the vectorized $\alpha_{i,s}$. A common approach for these types of problems is alternating minimization [17]: solving for one variable while fixing the other variable (iteratively). This amounts to solving the following subproblems

$$\hat{\boldsymbol{x}}^{[i+1]} = argmin_{\boldsymbol{x}} \left\| \boldsymbol{s} - \boldsymbol{BFC}\left(\hat{\boldsymbol{\theta}}^{[i]}\right)\boldsymbol{x} \right\|_2^2$$
$$\hat{\boldsymbol{\theta}}^{[i+1]} = argmin_{\boldsymbol{a}} \left\| \boldsymbol{s} - \boldsymbol{BFC}(\boldsymbol{\theta})\hat{\boldsymbol{x}}^{[i]} \right\|_2^2, \tag{2.44}$$

while fixing one variable at a time. The first problem is a simple linear inverse problem. Since $\boldsymbol{C}$ is block diagonal, the second problem can be rewritten as

$$\hat{\boldsymbol{\theta}}^{[i+1]} = argmin_{\boldsymbol{a}} \left\| \boldsymbol{s} - \boldsymbol{BF}\tilde{\boldsymbol{C}}(\hat{\boldsymbol{x}}^{[i]})\boldsymbol{\theta} \right\|_2^2, \tag{2.45}$$

where each row i of $\tilde{\boldsymbol{C}}(\boldsymbol{x})$ is given by $\left[r_i^{\langle 1 \rangle} x_i, r_i^{\langle 2 \rangle} x_i, \ldots, r_i^{\langle P \rangle} x_i\right]$. Under this formulation, estimation of $\boldsymbol{\theta}$ becomes a linear inverse problem as well. Further discussion of various (linear and nonlinear) methods for parallel imaging can be found in Chapter 6.

2.6.2 Nonlinear motion estimation/correction

MR acquisition times are typically in the order of minutes, making the scan sensitive to motion. Reconstructing data from a moving object will lead to artifacts similar to aliasing artifacts (in addition to blurring artifacts) because the reconstructed image will correspond to aliased views of the object in the various motion states. One way to correct for these artifacts is to incorporate motion into the forward model $\boldsymbol{E}$

$$\hat{\boldsymbol{x}} = argmin_{\boldsymbol{x}} \left\| \boldsymbol{s} - \boldsymbol{BFCMx} \right\|_2^2, \tag{2.46}$$

where $\boldsymbol{B} = [\boldsymbol{B}_1, \ldots, \boldsymbol{B}_n]^T$ is the sampling for each motion state n (logical, block diagonal sparse matrix) and $\boldsymbol{M} = [\boldsymbol{M}, \ldots, \boldsymbol{M}_n]^T$ are the corresponding motion fields (real sparse matrix). If the motion is known (along with coil sensitivities and motion state information), the motion corrected image can be easily obtained since this is a linear inverse problem (note that the problem of motion corrected reconstruction is generally ill-posed since rotation or scaling motion can open gaps in k-space, effectively inducing undersampling). A nonlinear problem arises when we attempt to simultaneously estimate the motion corrected image and the motion

$$\left\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{\theta}}\right\} = argmin_{\boldsymbol{x},\boldsymbol{\theta}} \left\| \boldsymbol{s} - \boldsymbol{BFCM}(\boldsymbol{\theta})\boldsymbol{x} \right\|_2^2, \tag{2.47}$$

where $\boldsymbol{\theta}$ are the motion parameters (we assume $\boldsymbol{B}$ is known). This problem is generally underdetermined due to the large number of unknowns, $K_{samples} < N_{pixels} + DN_{pixels}N_{shots}$, where $D = 2$ (3) for 2D (3D) imaging. If the motion can be captured by a simple model, e.g., rigid motion, then the conditioning of the problem can improve significantly [18]. Considering 2D rigid motion, we can reduce the number of motion parameters from $DN_{pixels}N_{shots}$ to $3N_{shots}$ since $\boldsymbol{\theta}_{(n)} = \left[t_{u(n)}, t_{v(n)}, \alpha_{(n)}\right]^T$, where $t_{u(n)}$ and $t_{v(n)}$ are global translations (along two dimensions, $\cdot_{u,v}$) and $\alpha_{(n)}$ is the rotation angle (for the n^{th} motion state). This can be a very useful motion model for brain imaging, for example. While the dependency on the translational component of motion is linear, it is not the case for the rotational component that involves $\sin(\alpha)$ and $\cos(\alpha)$. As seen in the previous section, one way to tackle joint estimation is to iteratively alternate between two subproblems

$$\begin{aligned} \hat{\boldsymbol{x}}^{[i+1]} &= argmin_{\boldsymbol{x}} \left\| \boldsymbol{s} - \boldsymbol{BFCM}(\hat{\boldsymbol{\theta}}^{[i]})\boldsymbol{x} \right\|_2^2 \\ \hat{\boldsymbol{\theta}}^{[i+1]} &= argmin_{\boldsymbol{\theta}} \left\| \boldsymbol{s} - \boldsymbol{BFCM}(\boldsymbol{\theta})\hat{\boldsymbol{x}}^{[i]} \right\|_2^2 . \end{aligned} \tag{2.48}$$

Once again, the first problem is a straightforward linear inverse problem. For the second problem, one approach is to employ the Newton method by performing a second-order Taylor expansion

$$\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]} + d\boldsymbol{\theta}\right) \approx \boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right) + \mathcal{D}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right) d\boldsymbol{\theta} + \frac{\mathbf{1}}{\mathbf{2}}\mathcal{H}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right) d\boldsymbol{\theta}^2, \tag{2.49}$$

where $\mathcal{H}$ is the Hessian. It can be shown that $d\boldsymbol{\theta} = -\left[\mathcal{H}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right)\right]^{-\mathbf{1}} \mathcal{D}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right)$, making the corresponding Newtown update as follows: $\boldsymbol{\theta}^{[i+1]} = \boldsymbol{\theta}^{[i]} - \left[\mathcal{H}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right)\right]^{-\mathbf{1}} \mathcal{D}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right)$. Since the Hessian is not guaranteed to be positive definite, not all iterations are guaranteed to reduce the cost function. The Levenberg–Marquardt is a common variant of the algorithm that ensures convergence at every step; in this case, the update is given by $\boldsymbol{\theta}^{[i+1]} = \boldsymbol{\theta}^{[i]} - \left[\mathcal{H}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right) + \alpha^{[i]}\boldsymbol{I}\right]^{-\mathbf{1}} \mathcal{D}\boldsymbol{E}\left(\boldsymbol{\theta}^{[i]}\right)$, with $\alpha \geq 0$. If a rigid (or affine) motion model is considered, then the first and second derivatives, $\mathcal{D}\boldsymbol{M}(\boldsymbol{\theta})$ and $\mathcal{H}\boldsymbol{M}(\boldsymbol{\theta})$, are practical to compute.

A more challenging problem arises when affine motion models are not sufficient to accurately describe the underlying motion (e.g., of a respiratory or cardiac nature). In this case, the motion is modeled as a deformation field ($\boldsymbol{w}$) with two parameters (with directions denoted as $\cdot_{u,v}$), per pixel ($.^{(x_i)}$), per motion state ($\cdot_{(n)}$), i.e., $\boldsymbol{\theta}_{(n)} = \left[w_{u(n)}^{x_1}, w_{v(n)}^{x_1}, \ldots, w_{u(n)}^{x_N}, w_{v(n)}^{x_N}\right]^T$. One approach to this problem is to establish a link between residual errors in the image $\boldsymbol{x}$ and small perturbations in the motion model $\boldsymbol{M}(\boldsymbol{\theta})$ [19]. In the presence of errors in the model (namely, in the motion operator), there will be a residual error in the reconstruction, $\boldsymbol{s} = \boldsymbol{E}(\hat{\boldsymbol{\theta}})\boldsymbol{x} + \mathcal{E}$, where $\mathcal{E} = \boldsymbol{BFC}\left[\boldsymbol{M}(\boldsymbol{\theta}_{true}) - \boldsymbol{M}(\hat{\boldsymbol{\theta}})\right]\boldsymbol{x}_{true}$, $\boldsymbol{x}_{true}$ is the motion free image and $\boldsymbol{\theta}_{true}$ are the correct motion parameters. A key assumption based on optical flow makes possible the following approximation: $\left[\boldsymbol{M}(\boldsymbol{\theta}_{true}) - \boldsymbol{M}(\hat{\boldsymbol{\theta}})\right]\boldsymbol{x}_{true} \cong -\nabla_s\hat{\boldsymbol{x}}\ d\boldsymbol{\theta}$, where ∇_s denotes the spatial gradient. $\nabla_s\hat{\boldsymbol{x}}$ is a block (for each motion state $\cdot_{(n)}$), bidiagonal matrix, where the i^{th} row of the n^{th} motion state is given by $\nabla_s\hat{\boldsymbol{x}}_{i(n)} = \left[0, \ldots, 0, \partial x_{i(n)}/\partial u, \partial x_{i(n)}/\partial v, 0, \ldots, 0\right]$. Under this approximation, we can establish the following relationship

$$\mathcal{E} = \boldsymbol{s} - \boldsymbol{BFCM}\left(\hat{\boldsymbol{\theta}}\right)\boldsymbol{x} = -\boldsymbol{BFC}\nabla_s\hat{\boldsymbol{x}}d\boldsymbol{\theta}. \tag{2.50}$$

Consequently, we can write two coupled problems to be solved by alternate minimization as

$$\begin{aligned} \hat{\boldsymbol{x}}^{[i+1]} &= argmin_{\boldsymbol{x}} \left\| \boldsymbol{s} - \boldsymbol{BFCM}(\hat{\boldsymbol{\theta}}^{[i]})\boldsymbol{x} \right\|_2^2 \\ d\hat{\boldsymbol{\theta}}^{[i+1]} &= argmin_{d\boldsymbol{\theta}} \left\| \boldsymbol{s} - \boldsymbol{BFCM}\left(\hat{\boldsymbol{\theta}}^{[i]}\right)\hat{\boldsymbol{x}}^{[i]} + \boldsymbol{BFC}\nabla_s\hat{\boldsymbol{x}}^{[i]}d\boldsymbol{\theta} \right\|_2^2 , \end{aligned} \tag{2.51}$$

with the motion model parameters being updated by $\boldsymbol{\theta}^{[i+1]} = \boldsymbol{\theta}^{[i]} + d\boldsymbol{\theta}^{[i]}$. Note that both problems in Eq. (2.51) are linear inverse problems and have practical solutions. The second problem, however, is highly underdetermined ($DN_{pixels}N_{shots}$ unknowns). Parametrization of the motion and further regularization can be employed to improve the condition of the second problem. The general problem of motion estimation and motion correction in MRI is discussed in greater detail in Chapter 13.

2.6.3 Nonlinear parameter reconstruction

One of the strengths of MRI is the capability of measuring a plethora of physical tissue and system properties, such as relaxation times, diffusion, susceptibility, temperature, water-fat separation, magnetization transfer, and magnetic field (B0 or B1) mapping. Conventional methods to estimate these parameters rely on a two-step approach: i) reconstruct a set of images with different contrasts, weighted by the parameter of interest; ii) perform a pixel-wise fit of the reconstructed images to the theoretical MR physics model to estimate the parameter of interest. Note that the reconstruction step in i) requires $K_{samples} = N_{contrasts}N_{pixels}$ to produce alias-free images, enabling the fit of a parametric map of N_{pixels}. On the other hand, reconstructing the parameter map directly from the acquired k-space data can reduce the number of unknowns, improving performance. Similar to previous examples, the MR physics parameters of interest will be embedded into the encoding operator, and their direct estimation will require challenging nonlinear reconstructions.

The initial forward model outlined at the start of the chapter (Eq. (2.1)) neglects the fact that the transverse magnetization M_{xy} is a function tissue parameters $\boldsymbol{\theta}$ such as the proton density ρ, relaxation times T_1, T_2 or T_2^* (and/or other MR physics parameters), and sequence parameters $\boldsymbol{\kappa}$ that control the RF excitation field $\mathbf{B_1}$ and the magnetic field gradients $\mathbf{G}$. In other words, $M_{xy}(\boldsymbol{r}, t) = Z(\boldsymbol{r}, t, \boldsymbol{B_1}, \boldsymbol{G}, \boldsymbol{\theta})$, where Z models the Bloch equations. For simplicity, we will consider steady-state applications, where the signal model is usually in the form $M_{xy}(\boldsymbol{r}) = Z_{\boldsymbol{\kappa}}(\boldsymbol{r}, \boldsymbol{\theta})\ \rho(\mathbf{r})$, where Z is the signal model for the MR sequence determined by sequence parameters $\boldsymbol{\kappa}$. Inserting our new signal model into Eq. (2.1) leads to the following inverse problem:

$$\left\{\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\theta}}\right\} = argmin_{\boldsymbol{\rho},\boldsymbol{\theta}} \|\boldsymbol{s} - \boldsymbol{F}\boldsymbol{C}\boldsymbol{Z}_{\boldsymbol{\kappa}}(\boldsymbol{\theta})\boldsymbol{\rho}\|_2^2 + \sum_i \lambda_i \boldsymbol{R_i}(.), \tag{2.52}$$

which is an instance of Eq. (2.39). Despite being linear on $\boldsymbol{\rho}$, this problem is not linear on $\boldsymbol{\theta}$. As shown in the previous sections, a Taylor expansion can be performed to linearize the problem around an existing solution [20]

$$\boldsymbol{E}\left(\boldsymbol{y}^{[i]} + d\boldsymbol{y}\right) \approx \boldsymbol{E}\left(\boldsymbol{y}^{[i]}\right) + \mathcal{D}\boldsymbol{E}\left(\boldsymbol{y}^{[i]}\right) d\boldsymbol{y} = \boldsymbol{s}, \tag{2.53}$$

with $\boldsymbol{E} = \boldsymbol{F}\boldsymbol{C}\boldsymbol{Z}_{\boldsymbol{\kappa}}(\boldsymbol{\theta})$ and $\boldsymbol{y} = [\boldsymbol{\rho}, \boldsymbol{\theta}]$. This leads to the same strategy outlined previously in nonlinear parallel imaging reconstruction and can be solved by some Gauss–Newton variant. As before, the initialization of $\boldsymbol{y}^{[0]}$ can determine the convergence of the algorithm. Moreover, the scales of the partial derivatives of $\boldsymbol{Z}_{\boldsymbol{\kappa}}$ can vary by several orders of magnitude (depending on the MR physics model and parameters under consideration), which can significantly increase the condition number of the problem. This issue is addressed by normalizing these partial derivatives (a form of preconditioning); the normalizing scales are often experimentally determined. If the partial derivatives are not normalized, the iterative process may take a very large number of iterations to converge or, possibly, diverge. Finally, a variety of regularization approaches can be enforced for any of the parameters being reconstructed; since the parameter maps often have similar structure to common contrast-weighted MR images, similar regularization strategies apply.

A similar approach [21] exploits the fact that, for a given $\boldsymbol{\theta}$, the linear term $\boldsymbol{\rho}$ can be directly obtained by solving the corresponding linear inverse problem $\boldsymbol{\rho} = [\boldsymbol{FCZ_{\kappa}}(\boldsymbol{\theta})]^{\boldsymbol{H}}\, \boldsymbol{s}$. Reinserting this into Eq. (2.52) leads to:

$$\hat{\boldsymbol{\theta}} = argmin_{\boldsymbol{\theta}} \left\| \left\{ \boldsymbol{I} - \boldsymbol{FCZ_{\kappa}}(\boldsymbol{\theta}) \left[\boldsymbol{FCZ_{\kappa}}(\boldsymbol{\theta})\right]^{\boldsymbol{H}} \right\} \boldsymbol{s} \right\|_2^2 + \sum_i \lambda_i \boldsymbol{R_i}(.). \tag{2.54}$$

Recasting the problem in this manner reduces the total number of unknowns and is expected to have better convergence properties (i.e., avoid local minima). The same strategy as outlined in Eq. (2.48) can be used here, solving the nonlinear problem with some Newton-variant algorithm. As seen in previous examples, these will require the Jacobian (and potentially the Hessian) of the encoding operator, which may incur considerable computation costs. Notably, in nonlinear parameter reconstruction, $\partial \boldsymbol{Z}_k(\boldsymbol{\theta})/\partial \boldsymbol{\theta}$ will need to be computed. For most simple cases, $\boldsymbol{Z}_k$ is an exponential of some sort, and an analytical solution exists. On the other hand, if a general (nonsteady state) MR sequence is considered, then $\boldsymbol{Z_{\kappa}}(\boldsymbol{\theta}, t)$ may not have a closed-form solution, requiring a full evaluation of the Bloch equations instead. In such cases, finite difference methods may be employed to approximate the required partial derivatives of the MR sequence forward model $\boldsymbol{Z_{\kappa}}(\boldsymbol{\theta}, t)$. Calculating the derivative via finite differences will require repeated evaluations of $\boldsymbol{Z_{\kappa}}(\boldsymbol{\theta}, t)$ and $\boldsymbol{Z_{\kappa}}(\boldsymbol{\theta} + \boldsymbol{\Delta\theta}, t)$, which can add to the computational burden, particularly as the complexity of the MR physics model $\boldsymbol{Z_{\kappa}}$ increases and, correspondingly, the number of parameters in $\boldsymbol{\theta}$. A thorough description of various linear and nonlinear approaches to parametric mapping is provided in Chapter 15.

2.7 Summary

In this chapter we covered the fundamentals of reconstruction in MRI, how it can be formulated as a linear inverse problem, and some of the tools we have available to solve this problem. MR reconstruction is generally an ill-posed problem (in the Hadamard sense), with several considerations that underpin the process. The required discretization and finite sampling during acquisition impose limitations on the reconstructed alias-free FOV, resolution, and potential *Gibbs-ringing* artifacts. The presence of Gaussian noise during the acquisition also dictates that the best image estimate will be given by the minimization of a L_2 data consistency term. In the case of a *fully sampled* acquisition, the process requires only an inverse Fourier transform; however, in the general *undersampled* case, the problem is underdetermined, and additional information is required. We explored various types of regularization for *undersampled* reconstructions, using L_1 and L_2 norms, and how one goes about solving these problems. In the last part of this chapter, we briefly introduced some nonlinear inverse problems in the context of MR reconstruction. These usually arise when the unknown is not (just) the image, but some parameters that nonlinearly characterize the forward model. Nonlinear inverse problems are present in many MR applications and have been tackled with a myriad of different strategies, but we introduced only a few in this chapter. In the remaining chapters of the book we will explore all of the key methods in MR reconstruction, from seminal to state-of-the-art, for linear and nonlinear inverse problems.

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